

Embedded Engineer Intern – Technical Task Requirements

Role: Embedded Engineer Intern

2. ESP32 Firebase Requirement

Task 9:

GPIO :

PUMP_LED 2

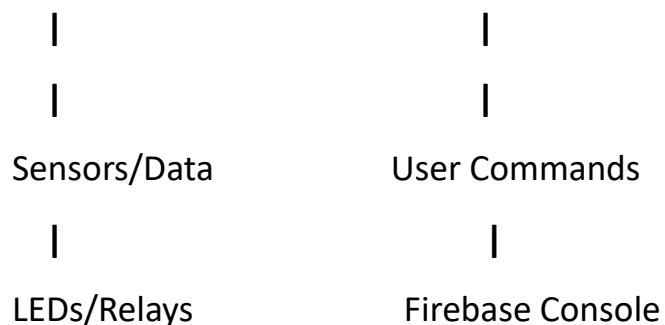
CHARGE_LED 4

ALARM_LED 5

BUZZER_LED 18

System Architecture

ESP32 →WiFi← FireBase Realtime Database



Monitoring Parameters :

solarVoltage,batteryVoltage,pumpCurrent,tankLevel,pumpStatus
,chargingStatus,dryRunStatus,systemStatus

Solar Voltage = 19.6V

Battery Voltage = 12.6V

Pump Current = 3.9A

Tank Level = 75%

>Currently, these values are simulated using random numbers.

>ESP32 uploads these values periodically to Firebase.

>Firebase acts as a cloud dashboard.

LED Simulation

Since actual hardware relays and pumps are not connected, LEDs simulate outputs.

For ex: Pump Control

pumpCommand=1

ESP32 turns ON Pump LED.

pumpCommand=0

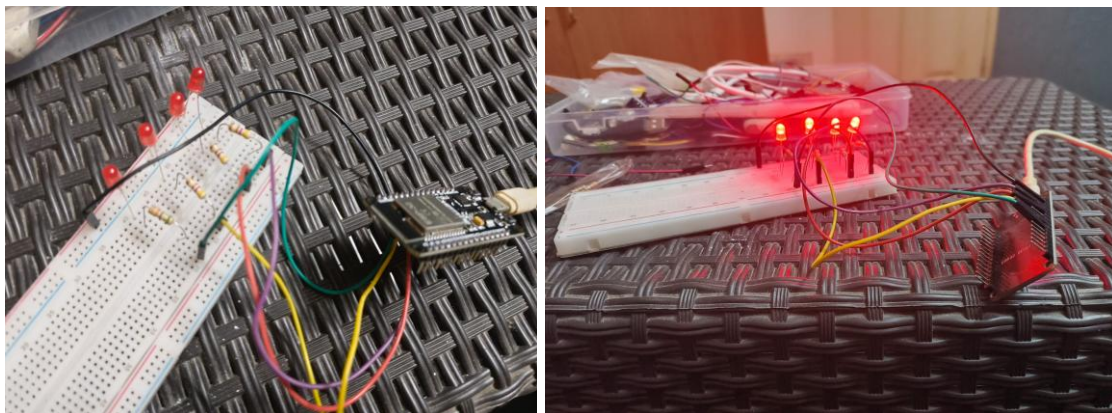
ESP32 turns OFF Pump LED.

This example goes for all four Sensor.

Project Workflow

1. ESP32 reads sensor values
2. ESP32 uploads data to Firebase
3. User views data remotely
4. User changes command in Firebase
5. ESP32 reads command
6. ESP32 activates relay/LED
7. Status updated back to Firebase

This project implements a cloud-based solar pump monitoring and control system using ESP32 and Firebase Realtime Database. The ESP32 uploads parameters such as solar voltage, battery voltage, pump current, tank level, and system status to Firebase. Users can remotely control the pump, charging relay, alarm, and buzzer through Firebase commands. LEDs are used to simulate relay outputs. The system demonstrates real-time IoT monitoring and remote control of a solar pumping application.



```
sketch_jun19c | Arduino IDE 2.3.5
File Edit Sketch Tools Help
DOIT ESP32 DEVKIT V1
sketch_jun19c.ino
128
Serial Monitor X
[message error to send message to 'DOIT ESP32 DEVKIT V1' on 'COM7']
connecting WiFi...
WiFi Connected
IP Address: 10.244.70.213
Firebase SignUp Success
Waiting for Firebase...
Firebase Ready
Data Uploaded
Pump Command = 0
Charging Command = 0
Alarm Command = 0
Buzzer Command = 0
Data Uploaded
Pump Command = 0
Charging Command = 1
Alarm Command = 0
Buzzer Command = 0
Data Uploaded
Pump Command = 0
Charging Command = 1
Alarm Command = 1
Buzzer Command = 0
Data Uploaded
Pump Command = 0
Charging Command = 1
Alarm Command = 1
Buzzer Command = 1
Data Uploaded
Pump Command = 1
Charging Command = 1
Alarm Command = 1
```